

Software Requirements Specification (SRS)

Project: ArcadeHub – Multiple games in one app (UNO, Puzzle & Memory Game System)

1. Introduction

The Multi-Game Platform is a web-based application that allows users to access and play multiple games such as UNO, Puzzle, and Memory Game within a single interface. Users must register and log in to access the system. The application provides an interactive gaming experience along with score tracking and leaderboard features. External APIs are integrated to enhance gameplay by providing dynamic content such as card decks and images.

2. Objectives

- Provide a centralized platform for multiple games
- Secure user login and authentication
- Enable multiplayer gameplay (UNO)
- Offer different difficulty levels (Memory Game)
- Maintain score tracking and leaderboard
- Ensure smooth and interactive user experience

3. Users

- Registered Users (Players)
- Administrator (optional for future enhancements)

4. Functional Requirements

User Management

- User registration and login
- Input validation for user data
- Session handling

Game Management

- Display list of available games (UNO, Puzzle, Memory)
- Start and manage game sessions
- End game and return results

UNO Game (Multiplayer)

- Create or join game room
- Distribute cards to players
- Turn-based gameplay
- Apply UNO rules (Skip, Reverse, Draw cards, etc.)
- Declare winner

Puzzle Game

- Load image (from API or static)
- Convert image into grid
- Shuffle tiles
- Allow user to rearrange tiles
- Detect correct arrangement

Memory Game

- Select difficulty level (Easy, Medium, Hard)
- Generate card grid based on difficulty
- Flip and match cards
- Track attempts and completion

Score & Leaderboard

- Calculate score based on game performance
- Store score in database
- Display leaderboard
- Show top players per game

External API Integration

- Deck of Cards API for UNO game (card generation and distribution)
- Image API for Puzzle game (dynamic images for gameplay)

5. Non-Functional Requirements

- Performance: Fast response time for gameplay actions
- Security: Secure handling of user data
- Usability: Simple and user-friendly interface
- Scalability: Ability to add more games in future
- Reliability: Stable system during gameplay sessions

6. Technology Stack

- Frontend: React (JavaScript)
- Backend: Spring Boot (Java)
- Database: PostgreSQL
- ORM: Spring Data JPA
- Security: Spring Security + JWT

7. Database Tables

- Users, Games, Scores, Game_Sessions

8. Future Enhancements

- Real-time multiplayer using WebSockets
- AI-based opponent for UNO
- Achievements and badges system
- Mobile application version